Electricity & Magnetism

Chapter one : Electrostatics

Coulomb's Law:

The electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. The force acts along the line joining the two charges.

Limitations of Coulomb's Law:

1. ***Only for Point Charges*:** Coulomb's law works accurately only when the charges are point-sized. If the charges have size or shape, the law becomes less accurate.
2. ***Valid in Electrostatics Only*:** It applies only to stationary (non-moving) charges. If the charges are moving, magnetic effects come in then Coulomb's law doesn't apply.
3. ***Works Best in Vacuum or Air*:** The law is accurate in vacuum or dry air.
4. ***Does Not Consider Other Forces*:** Coulomb's law only considers electric force, not other forces like gravity or magnetism.
5. ***Fails at Very Short Distances*:** The distance between the charges must be greater than the nuclear size. At very short distances, quantum effects become important, and Coulomb's law doesn't hold exactly**.**

Point Charge: A point charge is a charge that is considered to exist at a single point in space. It has no size, only charge.

Permittivity of Vacuum (ε_0): It tells us how easily electric field lines can pass through empty space (vacuum). It is a constant that tells us how much resistance the empty space (vacuum) gives to the formation of an electric field. It helps in calculating the force and field between charges in space.

Dipole: A dipole is made of two equal and opposite charges placed very close to each other. It behaves like a tiny magnet with a positive and a negative end.

Electric Dipole: An electric dipole is a pair of opposite electric charges (positive and negative) that are placed close together. It creates an electric field and is common in molecules like water.

Dipole Moment: The dipole moment shows how strong a dipole is. It depends on the amount of charge and the distance between the charges.

Formula: $\vec{p} = q \times \vec{d}$

Where: p = dipole moment, q = charge, d = distance between charges.

Electric Field: An electric field is a region around a charged object where other charges feel a force. It shows how strong the force is and in which direction it acts.

Formula: E=F/Q

Where: E = electric field, F = force, Q = charge

Electric Field Due to a Dipole:

An electric dipole is made of two opposite charges (positive and negative) placed very close to each other. This dipole creates an electric field in the space around it. It depends on:

1. The amount of charge in the dipole
2. The distance from the dipole
3. The direction in which you're measuring the field

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{r^3}$$

Where:

- E = electric field (strength)
- p = dipole moment = $q \times 2a$
  - q = charge
  - $2a$ = distance between the charges
- r = distance from the dipole to the point where you're measuring the field
- ε_0 = permittivity of vacuum

Here, the electric field from a dipole becomes weaker very quickly as we move farther because of $1/r^3$. The dipole moment p tells us how strong the dipole is. The direction of the electric field depends on the position — the electric field comes out of the positive charge and goes into the negative charge.

Electric Field Density:

Electric Field Density is more commonly referred to as Electric Flux Density. Electric flux density tells us how much electric field passes through a given area. It shows how concentrated the electric field lines are in a certain region.

Formula:

$$D = \varepsilon \cdot E$$

Where:

- D = Electric flux density (C/m^2)
- ε = Permittivity of the medium
- E = Electric field (N/C)

When more electric field lines pass through a surface, the flux density becomes higher. It depends on the medium because different materials allow different amounts of field lines. It helps in understanding how materials react to electric fields.

Torque on a dipole in an electric field:

Let us consider an electric dipole consisting of two opposite charges of equal magnitudes +q and –q, the distance between them d is an external electric field $\vec{E}$, which experiences a dipole moment $\vec{p}$ making an angle θ with this field $\vec{E}$.

$$|\vec{F}|=|-\vec{F}|=q\vec{E}.$$

Since two forces are equal in magnitudes and opposite in direction, so they constitute a torque.

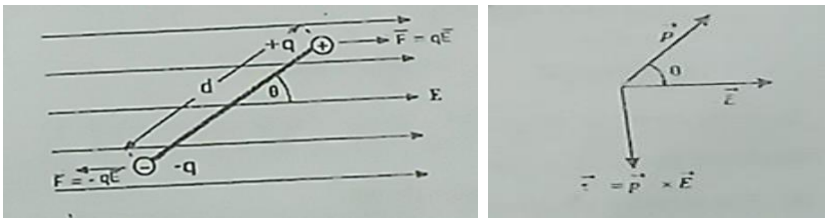


Fig: Dipole in an uniform electric field.

Now, τ= force × perpendicular distance

=F × dsinθ

=qEdsinθ [$\because \vec{F}$ = q$\vec{E}$]

=pEsinθ [$\because \vec{p}$= q$\vec{d}$]

As the direction of torque $\vec{\tau}$ is perpendicular with both $\vec{p}$ and $\vec{E}$, so we can write in vector form,

$$\vec{\tau} = \vec{p} \times \vec{E}$$

To change the orientation of the dipole from 90° to some other angle θ in the direction of an electric field, work must be done by an external agent. This work is stored as potential energy in the system of the dipole and the electric field.

If the dipole starts at 90° to the field and it points in the same direction as the field, turned to 0°, then the work done is:

$$U=W=\int_0^W dW$$

$$=\int_{90^\circ}^{\theta} \tau d\theta$$

$$=\int_{90^\circ}^{0} pE sin\theta d\theta$$

$$=pE \int_{90^\circ}^{0} sin\theta d\theta$$

$$=pE\ [-\cos\theta]_{90^\circ}^{\theta}$$

$$=-pE\ (\cos\theta - \cos 90^\circ)$$

$$=-pE\cos\theta$$

In vector form, U= $-\vec{p}.\vec{E}$

Special cases:

1. The torque is maximum when θ=90°.

Maximum value of torque, τ_{max}= pEsinθ = pEsin90° = pE.

2. When θ=0°, then τ=0.

An electric dipole is most stable when it points in the same direction as the electric field. In this position it has least potential energy and it won't try to move or rotate.

3. When θ=180°, then τ= pEsinθ = pEsin180° = 0.

The electric dipole is in an unstable equilibrium when it is held in a direction opposite to the direction of electric field.

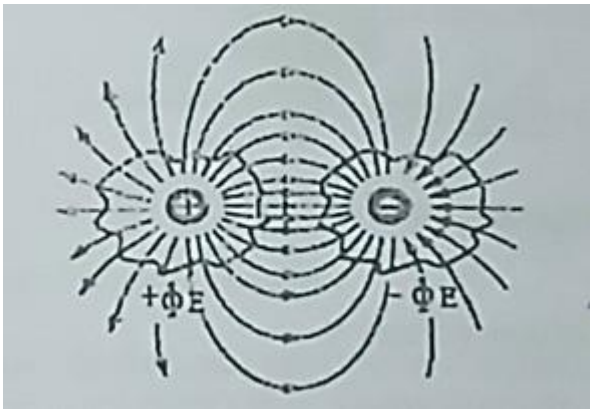
Summary:

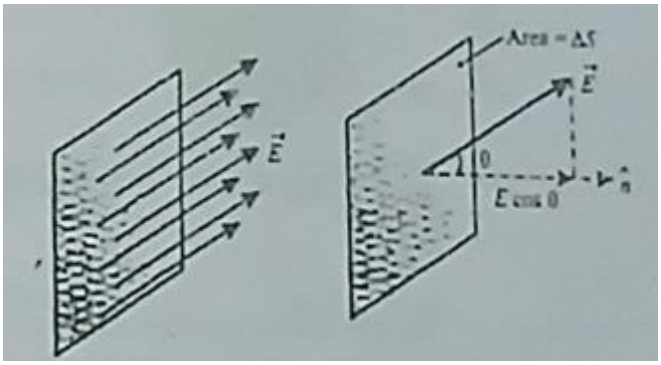
1. Torque is zero at both 0° and 180°.
2. But stability depends on potential energy.
3. 0°; minimum potential energy; -pE; stable.
4. 180°; maximum potential energy; +pE; unstable.

Electric Flux:

Electric flux is the amount of electric field passing through a surface. It shows how strong and how many electric field lines go through that surface. For an electric field, the flux *(ϕ)* is measured by the number of lines of force that pass through the surface, open or closed. For a closed surface:

- ϕ is positive if the field lines go outward from the surface.
- ϕ is negative if the field lines go inward toward the surface.





Let $d\phi$ be the number of lines cutting through dS. Then, E=$\frac{d\phi}{dS}$

⇨ $d\phi$ = E dS

⇨ $d\phi = \mathrm{E}\, dS\, cos\theta$

⇨ $d\phi = \vec{E}.\mathrm{d}\vec{S}$

Gauss's Theorem:

Gauss's Theorem states that the total electric flux passing through a closed surface depends only on the total charge inside that surface. It is equal to the total charge (Q) divided by the permittivity of the medium (ε). In formula:

$$\oint d\phi = \frac{Q}{\varepsilon}$$

⇨ $\varepsilon_0 \oint d\phi = Q$

⇨ $\varepsilon_0 \int \vec{\mathrm{E}}.\mathrm{d}\vec{\mathrm{S}} = \mathrm{Q}$ [For an open surface]

⇨ $\varepsilon_0 \oint \vec{E}.d\vec{S}$=Q [For an closed surface]

⇨ $\varepsilon_0 \sum \vec{\mathrm{E}}.\mathrm{d}\vec{\mathrm{S}} = \mathrm{Q}$ [For an irregular surface]

Here, ϕ= total electric flux, Q = total charge inside the surface, ε_0 = permittivity of the medium.

Proof of Gauss's Theorem:

Gauss's theorem states that "the total normal induction over a closed surface of any shape, drowns in an electric field is equal to 4 times the charge within the surface". If there is no charge inside, the normal induction over the surface is zero. The total electric flux through a closed surface is equal to the charge enclosed divided by the permittivity of free space:

$$\oint E.dS = \frac{q}{\varepsilon_0}$$

Let us consider a point charge q located inside a closed surface S and let us take a small surface element $d\vec{S}$ on the surface. The electric field $\vec{E}$ due to the point charge makes an angle θ with the outward normal to $d\vec{S}$. If $\hat{n}$ is a unit vector in the outward normal to the surface and $\hat{r}$ is a unit vector in the direction of $\vec{E}$, then

$$\oint E\,.dS = \oint E\hat{r}.\,\hat{n}\,\mathrm{dS}$$

Where $\vec{E}$ and $d\vec{S}$ are the magnitude of field intensity and the area element respectively.

Again, $E=\frac{1}{4\pi\varepsilon_0}.\frac{q}{r^2}$

$$\therefore \oint \vec{E}.d\vec{S}=\oint \frac{1}{4\pi\varepsilon_0}.\frac{q}{r^2}\ \hat{r}.\hat{n}\ dS$$

$$\Rightarrow \oint \vec{E}.d\vec{S} = \frac{q}{4\pi\varepsilon_0}\oint \frac{\hat{r}.\hat{n}}{r^2}dS$$

$$\Rightarrow \oint \vec{E}.d\vec{S} = \frac{q}{4\pi\varepsilon_0}\oint \frac{\cos\theta}{r^2}\ dS \quad [\because\ \hat{r}.\hat{n} = (1)(1)cos\theta = cos\theta]$$

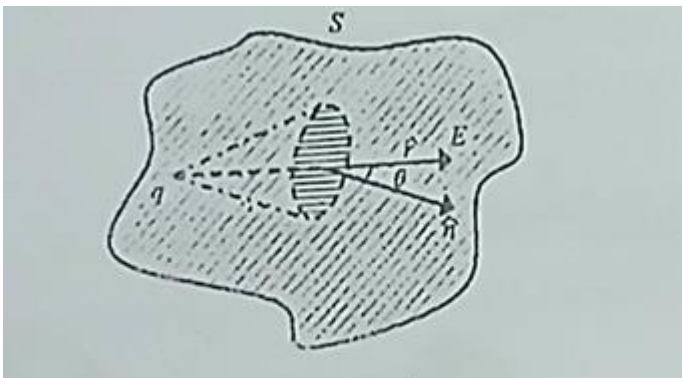


Fig.4.5 Gauss's law.

We know that, a solid angle is the ratio of the curved surface area on a sphere, enclosed by a cone to the square of the sphere's radius.

So, $\frac{\mathrm{dS}\cos\theta}{r^2}$ is the element of the solid angle dΩ subtended by the surface element dS at the position of q.

$$\because \oint \vec{E}.d\vec{S} = \frac{q}{4\pi\varepsilon_0}\oint d\Omega$$

But, $\oint d\Omega = 4\pi$

$\therefore \oint \vec{E}.d\vec{S} = \frac{q}{4\pi\varepsilon_0} 4\pi = \frac{q}{\varepsilon_0}$

Hence, Gauss's theorem is proved.

Applications of Gauss's Law:

1. If we know the strength of the electric field at every point on a closed surface, Gauss's Law helps us find the total charge inside that surface.
2. On the other hand, if we know how much charge is inside, Gauss's Law helps us find the electric field around it.
3. It is most useful when the shape of the system is flat (planar), round like a cylinder, or a sphere because of the symmetry.

<u>For bigger answer:</u>

1. It helps us to find the electric field around a single point charge.
2. It shows that the electric field inside a charged shell is zero.
3. It helps to calculate the electric field around a very long charged wire.
4. It gives a constant electric field near a large uniformly charged sheet.
5. It helps us to derive the formula for force between two charges.
6. It helps to check if any charge is present inside a closed surface.

Deduction of Coulomb's Law from Gauss's Law:

Coulomb's law can be derived from Gauss's law by using symmetry arguments. Let us consider a spherical surface of radius r, centered on a point charge q, as the Gaussian surface. Due to the uniform shape of the surface around the charge, the electric field $\vec{E}$ must point directly outward or inward if q is negative and must have the same magnitude at every point on the surface.

Since both $\vec{E}$ and the small area element $d\vec{S}$ point in the same direction at every point on this surface, the angle between them is zero. So Gauss's law becomes:

$$\varepsilon_0 \oint \vec{E}.d\vec{S} = q$$

$$\Rightarrow \varepsilon_0 \oint E dS cos\theta = q$$

$$\Rightarrow \varepsilon_0 \oint E dS = q$$

$$\Rightarrow \varepsilon_0 E(4\pi r^2) = q$$

$$\Rightarrow E = \frac{q}{4\pi r^2} \frac{1}{\varepsilon_0} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

The above equation gives the magnitude of the electric field $\vec{E}$ at any point at a distance r from an isolated point charge q.

Let us now put a second point charge q_0 at the point at which $\vec{E}$ is calculated. The magnitude of the force acting on it is,

$$\vec{F} = \vec{E}\, q_0$$

$$\Rightarrow \vec{F} = \frac{q q_0}{4\pi\varepsilon_0}\,\frac{1}{r^2}$$

This is precisely Coulomb's law.

Gauss's Law with a Dielectric:

Let us apply Gauss's law to parallel-plate capacitor filled with a dielectric of dielectric constant k. Let us consider a capacitor with and without the dielectric,

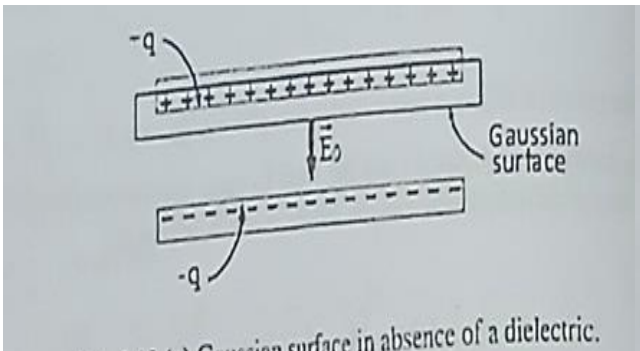


Fig. 5.12 (a) Gaussian surface in absence of a dielectric.

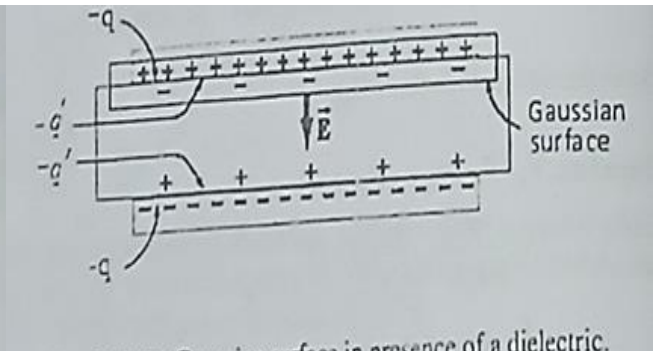


Fig. 5.12 (b) Gaussian surface in presence of a dielectric.

If no dielectric present, Gauss's law gives,

$$\varepsilon_0 \oint \vec{E}.d\vec{S} = q \ldots\ldots\ldots\ldots (1)$$

$$\Rightarrow \varepsilon_0 \oint E\,dS = q$$

$$\Rightarrow \varepsilon_0 EA = q$$

$$\Rightarrow E = \frac{q}{\varepsilon_0 A} \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

If dielectric is present, Gauss's law gives,

$$\varepsilon_0 \oint \vec{E}.d\vec{S} = q - q_1 \ldots\ldots\ldots\ldots\ldots(3)$$

$$\Rightarrow \varepsilon_0 EA = q - q_1$$

$$\Rightarrow E = \frac{q}{\varepsilon_0 A} - \frac{q}{\varepsilon_0 A} \ldots\ldots\ldots\ldots\ldots (4)$$

Where q_1 be the induced surface charge, so called free charge on the plates. The net charge within the Gaussian surface is $(q-q_1)$

The induced electric field, $E = \frac{E_0}{k} \ldots\ldots\ldots\ldots\ldots(5)$

Putting the value of E_0 in equation (5), we have

$$E = \frac{E_0}{k} = \frac{q}{\varepsilon_0 A}$$

Now inserting this in equation (4)

$$\Rightarrow \frac{q}{k\varepsilon_0 A} = \frac{q}{\varepsilon_0 A} - \frac{q_1}{\varepsilon_0 A}$$

$$\Rightarrow \frac{q}{k} = (q - q_1)$$

$$\Rightarrow q = qk - q_1 k$$

$$\Rightarrow q_1 k = qk - q$$

$$\Rightarrow q_1 = q\left(1 - \frac{1}{k}\right)$$

this result indicates that the induced surface charges q_1 is always less in magnitude than the free charge q and is equal to zero if no dielectric is present, that is if k = 1.

Substituting the value of q_1 in equation (3) ,

$$\varepsilon_0 \oint \vec{E}.d\vec{S} = q - q\left(1 - \frac{1}{k}\right)$$

$$\Rightarrow \varepsilon_0 k \oint \vec{E}.d\vec{S} = q$$

Capacitance and capacitor:

A capacitor is a passive device that stores charge and potential energy. Capacitance is the ability of a capacitor to store electric charge per unit voltage. The capacitance of a capacitor is the ratio of the charge stored in the capacitor plates to the potential difference between them. This consists of two conducting parallel plates of equal shape, carrying opposite charges and separated by a distance ‘d’ with a dielectric. Normally, polystyrene, oil, air and some ceramics are used as dielectric in some capacitor.

There are many kinds of capacitor. Such as:

1. Paper capacitor.
2. Mica capacitor.
3. Variable capacitor.
4. Electrolytic capacitor. Etc.

Uses of capacitor:

1. To establish strong fields within a small space.
2. To store electrical energy and use it at intervals.
3. The capacitor blocks the flow of direct current and allows the flow of alternating current.
4. To reduce voltage fluctuation and to provide time delays.
5. Capacitors are used in radio and TV receivers and transmitter circuits to create repeating signals, to connect different parts of the circuit.

Capacitance:

The charge 'Q' stored on either plate is proportional to the potential difference 'V' between the plates. The proportionally constant is called the capacitance.

$$\therefore Q \propto V. \qquad Q = CV.$$

$$\text{Capacitance, } C = \frac{Q}{V} = \frac{coulomb}{volt} = F(\text{farad})$$

Energy of a charged capacitor:

Let us consider an uncharged capacitor is being charged by gradually increasing the potential difference from 0 to V. the charge will increase from 0 to Q. let q and V be the charge and potential of the capacitor in the process of charging. Then the total work done is,

$$W = \int dW$$

$$\Rightarrow \int_0^W dW = \int_0^Q V dq$$

$$\Rightarrow \int_0^W dW = \int_0^Q \frac{q}{C} \qquad [\text{ as } C = \frac{q}{V}]$$

$$\Rightarrow \int_0^W dW = \frac{1}{C} \int_0^Q q$$

$$\Rightarrow \int_0^W dW = \frac{1}{C} \left[\frac{q^2}{2}\right]_0^Q$$

$$\therefore W = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} CV^2 \qquad [\text{as } Q = CV]$$

The work done is stored up in the form of potential energy and supplies the energy as necessary to drive the current when the capacitor is discharged.

Capacitor with a dielectric:

A capacitor with a dielectric is a device that stores electric charge. It has two metal plates with an insulating material (called a dielectric) placed between them. The dielectric helps the capacitor store more charge by increasing its ability to hold electricity.

1. Capacitance of a Parallel Plate Capacitor (No Dielectric)

$$C_0 = \frac{\varepsilon_0 A}{d}$$

- C_0: Capacitance without dielectric
- ε_0: Vacuum permittivity (~8.85 × 10^{-12} F/m)
- A: Area of each plate
- d: Distance between plates

2. With a Dielectric Inserted

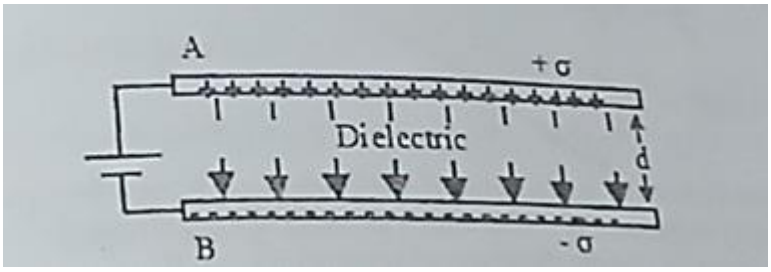
$$C = \kappa C_0 = \frac{\kappa \varepsilon_0 A}{d}$$

- κ: **Dielectric constant** (also called relative permittivity)
- C: New capacitance with dielectric
- $\kappa > 1$ means increased capacitance

Parallel Plate Capacitor:

A parallel plate capacitor is made of two flat metal plates placed parallel to each other, with a small space between them. The space may be filled with air or a dielectric. It stores electric charge and energy.

Let us consider a parallel plate capacitor which consists of two metal plates A and B separated by a distance d with a dielectric as shown in Figure. Let + σ be the surface charge density of plate A and – σ be the induced on plate B.



From Gauss's law, we have

$$\varepsilon_0 \oint \vec{E}.d\vec{S} = q$$

⇨ $\varepsilon_0 \oint E dS = q$

⇨ $\varepsilon_0 E \oint dS = q$

⇨ $\varepsilon_0 EA = q$

⇨ $E = \frac{q}{\varepsilon_0 A}$

⇨ $E = \frac{\sigma}{\varepsilon_0}$

Since, $\sigma = \frac{q}{A}$

The potential difference between the two plates is $V = V_1 - V_2$

Hence, V=Ed

$$\Rightarrow V = \frac{\sigma d}{\varepsilon_0}$$

The charge density,

$$\sigma = \frac{q}{A}, \text{ or } q = \sigma A$$

$$\therefore V = \frac{qd}{A\varepsilon_0}$$

$$\therefore \frac{q}{V} = \frac{\varepsilon_0 A}{d}$$

$$\therefore C = \frac{\varepsilon_0 A}{d}$$

Spherical Capacitor:

A spherical capacitor has two concentric spherical conductors, one inside the other. The space between them may be filled with air or a dielectric. It stores charge between the inner and outer spheres.

Formula:

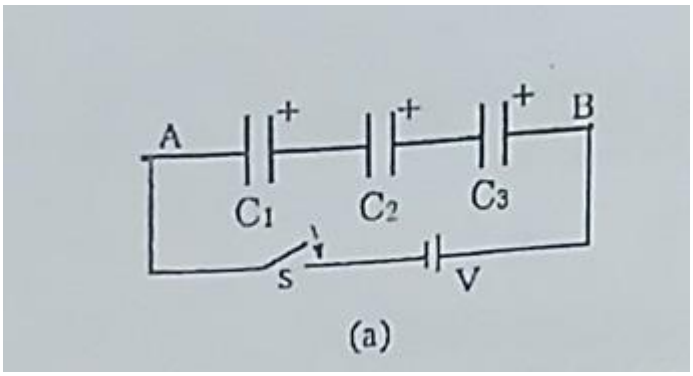
$$C = 4\pi\varepsilon_0 \frac{R_1 R_2}{R_2 - R_1}$$

Where:

- C = Capacitance
- ε_0 = Permittivity of free space
- R_1 = Radius of inner sphere
- R_2 = Radius of outer sphere

Equivalence Capacitance:

Series connection: Let us consider three capacitors C_1, C_2 and C_3 connected in series between the terminals A and B as shown in Fig: 1(a), when the switch S is closed the capacitors initially uncharged, becomes charged as the battery supplies an amount of charge Q.



By electrostatic induction equal and opposite charge accumulates on the opposite plates of the capacitors. If the potential difference across the capacitor is V,

Then $V = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3} = Q\left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right]$

Then $V = \frac{Q}{C}$

$$\therefore \frac{Q}{C} = Q\left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right]$$

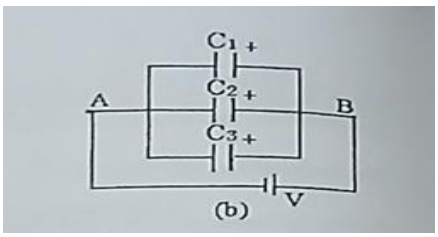
$$\therefore \frac{1}{C} = \left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right]$$

We can extend this argument to the case of N capacitor in series, with the result,

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \text{ ---------- } + \frac{1}{C_N}$$

Parallel connection: Let us consider three capacitors C_1, C_2 and C_3 connected in parallel between the terminals A and B as shown in Fig: 1(b), when the switch S is closed the capacitors initially uncharged, becomes charged as the battery supplies an amount of charge Q. this charge is shared by the three capacitors, so that

$$Q = Q_1 + Q_2 + Q_3$$



The potential differences are both equal to V; thus, we have

$$\Rightarrow Q = Q_1 + Q_2 + Q_3 = VC_1 + VC_2 + VC_3 = V(C_1 + C_2 + C_3)$$
$$\Rightarrow \frac{Q}{V} = C_1 + C_2 + C_3$$

If some charge Q supplies to the equivalent capacitor C, then

$$C = C_1 + C_2 + C_3 \qquad\qquad Q = CV$$

Or in general for N capacitors,

$$C = C_1 + C_2 + C_3 + \ldots\ldots\ldots\ldots\ldots\ldots + C_N$$